Brent Pappas

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Email: brent.pappas@ucf.edu
Portfolio: http://www.pappasbrent.com
GitHub: https://www.github.com/PappasBrent

Education

PhD. Computer Science. GPA: 3.983. University of Central Florida. Orlando, Florida. August 2021 - Present.

Master of Science. Computer Science. GPA: 3.983. University of Central Florida. Orlando, Florida. August 2021 - December 2025.

Bachelor of Science. Computer Science. GPA: 3.849. University of Central Florida. Orlando, Florida. August 2017 - May 2021.

Teaching

Fall 2026	**Instructor of Record.** COP-3402 Systems Software. UCF. Course homepage. Lecture Recordings.
Summer 2026	**Instructor of Record.** COP-3402 Systems Software. UCF. Course homepage. Lecture Recordings. Mean average teaching evaluation score: **4.72/5**. 81/107 enrolled students responded. Department mean was 4.29, college mean 4.23. Student ratings. Student comments.
Spring 2026	**Instructor of Record.** COP-3402 Systems Software. UCF. Course homepage. Lecture Recordings. Mean average teaching evaluation score: **4.62/5**. 164/250 enrolled students responded. Department mean was 4.24, college mean 4.21. Student ratings. Student comments.
Spring 2025	**Graduate Teaching Assistant.** COP-3402 Systems Software. UCF. Mean average overall teaching evaluation score: **4.93/5**. 15 students responded. Student perception of instruction.
Fall 2024	**Graduate Teaching Assistant.** COP-3402 Systems Software. UCF.
Fall 2019	**Peer Tutor Mentor.** COP-3502 Computer Science 1 & COP-3330 Object Oriented Programming. UCF.
Fall 2019	**Peer Tutor.** COP-3502 Computer Science 1 & COP-3330 Object Oriented Programming. UCF.
Spring 2019	**Peer Tutor.** COP-3502 Computer Science 1 & COP-3330 Object Oriented Programming. UCF.
Fall 2018	**Peer Tutor.** COP-3502 Computer Science 1. UCF.

Honors and Awards

2026	**GTA Excellence Award (University level)**. For providing exceptional teaching assistance to the course Systems Software. Awarded to 1 out of 800+ graduate teaching assistants. Award letter.
2026	**GTA Excellence Award (College level)**. For providing exceptional teaching assistance to the course Systems Software. Award letter.
2026	**CRA-WP 2026 Grad Cohort Workshop Invitee**. The CRA-WP arranged and paid for me to travel to the 2026 CRA-WP Grad Cohort Workshop in Seattle, Washington, because of my superior workshop application.
2024	**UCF 3MT Finalist**. For effectively communicating my research to a general audience in a 3 minute talk. Video. Certificate.
2024	**Distinguished Artifact Reviewer**. For my attention to detail while reviewing artifacts for OOPLSA 2024. Link.
2024	**FCI Student Scholarship**. For my interest in interdisciplinary research, scholarship, and collaboration. $5000 award.
2021	**Cum laude honors**. For having a cumulative undergraduate GPA greater than 3.8.
2017 - 2021	**Deans' list**. For continued undergraduate scholastic achievement.
2017	**Burnett Honors Scholar**. For academic talent, motivation, intellectual curiosity and creativity.
2017	**Provost Scholarship**. For outstanding academic performance in high school. Completely covered my undergraduate tuition fees at the University of Central Florida.

Publications

2026 **Translation Tag Team: Formal Rules and LLMs Translate More Macros Together than Apart**: by Brent Pappas, Joseph Zalusky, Zachary Burkett, and Paul Gazzillo. ASE. Acceptance rate of 20.20% (263 / 1304). Pre-print. Artifact. Badges: Available, functional, and reusable. Presentation. Slides.

2026 **Build Code is Still Code: Finding the Antidote for Pipeline Poisoning**: by Brent Pappas and Paul Gazzillo. ICSE NIER. Acceptance rate of 33.11% (50 / 151). Pre-print. Artifact. Badges: Available and reusable. Presentation. Slides.

2024 **Semantic Analysis of Macro Usage for Portability**: by Brent Pappas and Paul Gazzillo. ICSE. Acceptance rate of 22.26% (234 / 1051). Pre-print. Artifact. Badges: Available and reusable. Presentation. Slides.

2023 **Holy Macroni! A recipe for progressive language enhancement**: by Brent Pappas. Blog post for Trail of Bits. Blog post. Artifact.

2022 **SugarC: Scalable Desugaring of Real-World Preprocessor Usage into Pure C**: by Zach Patterson, Zenong Zhang, Brent Pappas, Shiyi Wei, and Paul Gazzillo. ICSE. Acceptance rate of 28.51% (197 / 691). Pre-print. Artifact. Badges: Available and reusable. Presentation.

Service

July 2026 **SPLASH/ISTTA 2026 Tool Demo Track Committee Member.** Virtual.

June 2026 **Camp Connect Lab Leader.** Orlando, Florida.

March 2026 **NAC - NAPC Volunteer**. Orlando, Florida.

October 2025 **UCF CS Grad Symposium 2025 Student Panelist**. Orlando, Florida.

August 2025 **UCF GradLaunch PhD Panelist**. Orlando, Florida.

August 2025 **UCF "Surviving Graduate School" Panelist**. Orlando, Florida.

June 2025 **Camp Connect Lab Leader**. Orlando, Florida.

May 2025 **UCF Graduate School Info Session PhD Panelist**. Orlando, Florida.

April 2024 **UCF CECS Senior Design Judge**. Orlando, Florida.

October 2022 **UCF CECS Career Kickoff Alumni Mentor**. Virtual.

May 2024 Orlando, Florida.

2024 **OOPLSA 2024 Artifact Evaluation Committee Member**. Virtual.

June 2023 **Camp Connect Lab Leader**. Orlando, Florida.

Nov - Dec 2022 **EuroSys 2023 Shadow Program Committee Member**. Virtual.

June 2021 **Camp Connect Lab Leader**. Orlando, Florida.

Presentations

2026 **ASE 2026: Translation Tag Team: Formal Rules and LLMs Translate More Macros Together than Apart.** Presentation at the 2026 IEEE/ACM International Conference on Automated Software Engineering YouTube recording. Slides.

2026 **Build Code is Still Code: Finding the Antidote for Pipeline Poisoning.** Presentation at the 2026 International Conference on Software Engineering in Rio de Janeiro, Brazil. YouTube recording. Slides.

2026 **Curing Pipeline Poisoning.** 3 minute thesis on my research on securing build systems against pipeline poisoning with build phase isolation. Presented at the CRA 2026 Grad Cohort Workshop. Recording link.

2026 **Invited Talk - The PhD Life.** Guest lecture on graduate student life presented to Sonoma State University's 2026 Spring Colloquium, hosted by Dr. John Sullins. Slides.

2025 **Invited Talk - The PhD Life.** Guest lecture on graduate student life presented to Dr. Suzanne Rivoire's undergraduate course, Computing Professions, at Sonoma State University. Slides.

2024 **Semantic Analysis of Macro Usage for Portability (ICSE 2024).** Presentation at the 2024 International Conference on Software Engineering in Lisbon, Portugal. Recording.

2023 **Semantic Analysis of Macro Usage for Portability (CAE-R)** Virtual presentation for Center of Academic Excellence in Cyber Research (CAE-R). Recording; jump to 25:38.

Posters

2026 Curing Pipeline Poisoning with Build Phase Isolation. 2026 Academy of Science, Engineering & Medicine of Florida Annual Meeting. Acceptance rate of 55.56% (150 / 270).

2024 Holy Macroni! Finding Macro Usage Errors in the Linux Kernel. 2024 International Symposium on Secure and Private Execution Environment Design. Link.

2022 Cpp2C: Transforming C Preprocessor Macros to C Language Constructs. 2022 University of Central Florida Student Scholar Symposium. Link.

Certifications

Completed higher education training, Preparing Tomorrow's Faculty

Level 1 College Readiness and Learning Assessment (CRLA) certified tutor.

Level 2 College Readiness and Learning Assessment (CRLA) certified tutor.

UCF Student Enhancement Training (SET) program certified.

Workshops

March 2026 2026 CRA-WP Grad Cohort Workshop. Seattle, Washington.

Oct - Nov 2025 2025 CRA-WP Virtual Mentoring Series Workshops. Virtual.

May 2025 2025 CRA-E Career Landscape Workshop. Virtual.

June 2022 Programming Language Mentoring Workshop (PLMW). San Diego, California.

Complete Work Experience

Jan 2026 - Present **Graduate Teaching Associate (Instructor of Record) for COP-3402 Systems Software**
University of Central Florida. Orlando, Florida.

May 2025 - Dec 2025 **Graduate Research Assistant**
University of Central Florida. Orlando, Florida.

Aug 2024 - May 2025 **Graduate Teaching Assistant for COP-3402 Systems Software**
University of Central Florida. Orlando, Florida.

April 2024 - July 2024 **Software Engineering Intern**
Trail of Bits. Remote.

Jan 2024 - April 2024 **Graduate Research Assistant**
University of Central Florida. Orlando, Florida.

Dec 2023 - Jan 2024 **Software Engineering Intern**
Trail of Bits. Remote.

Aug 2023 - Dec 2023 **Graduate Research Assistant**
University of Central Florida. Orlando, Florida.

May 2023 - Aug 2023 **Software Engineering Intern**
Trail of Bits. Remote.

Jul 2021 - May 2023 **Graduate Research Assistant**
University of Central Florida. Orlando, Florida.

Sep 2019 - Jul 2021 **Quality Assurance Coordinator**
SP+. Orlando, Florida.

Aug 2019 - Dec 2019 **Peer Tutor Mentor**
Student Academic Resource Center at the University of Central Florida. Orlando, Florida.

Aug 2018 - Dec 2019 **Peer Tutor**
Student Academic Resource Center at the University of Central Florida. Orlando, Florida.

Advising

Undergraduate Researchers

Aaron Burgess. 2026. Human factors in software engineering.

Thiago Airoldi. 2026. Analysis and security of software development pipelines.

Rasmin Ahmed. 2026. Analysis and security of software development pipelines.

Daniel Henriquez. 2026. Analysis and security of software development pipelines.

Brayden Coggin. 2026. Analysis and security of software development pipelines.

Thai Nguyen. 2025 - 2026. Human factors in software engineering and analysis and security of software development pipelines.

Nikola Krstajic. 2025. Human factors in software engineering.

Michael Johnson. 2025. Human factors in software engineering and analysis and security of software development pipelines.

Layla Le. 2025. Human factors in software engineering.

Kenneth Valladares. 2025. Human factors in software engineering.

Julian Velez. 2025. Human factors in software engineering.

Jorge Toledo. 2025. Human factors in software engineering.

Hasini Daliboyina. 2025. Human factors in software engineering.

David Gusmao. 2025 - 2026. Human factors in software engineering and analysis and security of software development pipelines.

Anju Thomas. 2025. Human factors in software engineering.

Adam Betinsky. 2025-2026. Human factors in software engineering and analysis and security of software development pipelines.

Ehigie Ekata. 2024. Semantics-aware translation of C preprocessor.

Zachary Burkett. 2024. Semantics-aware translation of C preprocessor.

Joseph Zalusky. 2024. Semantics-aware translation of C preprocessor macros.

Projects

2026 **Foreman**. University of Central Florida. A Python tool that enables developer to specify the commands file access permissions of phase of a project's build system, to protect these phases against pipeline poisoning attacks. Foreman is the tool associated with 2026 ICSE NIER paper, *Build Code is Still Code: Finding the Antidote for Pipeline Poisoning*.

2024 **Macroni**. Trail of Bits. A C static analysis tool that combines Trail of Bits' tools PASTA and VAST to lower C preprocessor macros down to MLIR. GitHub.

2024 **Maki**. University of Central Florida. A C++ Clang plugin that analyzes hows C preprocessor macros affect the C AST. Maki provides language porting tools information they can use to port macros to target languages in a way that preserves not just macro behavior, but also macro abstractions. Maki is the tool associated with 2024 ICSE paper, *Semantic Analysis of Macro Usage for Portability*, for which I am the lead-author. GitHub.

Skills Summary

Languages: Python3, C/C++, Bash, and HTML/CSS/JavaScript

Tools: Git, GitHub, and LLVM/Clang

Platforms: Linux, Web, and Windows

Soft Skills: Leadership, Writing, Communication, and Time management

References

Paul Gazzillo. Associate Professor. University of Central Florida. paul.gazzillo@ucf.edu. 407-823-5239. PhD supervisor.

Kevin Moran. Assistant Professor. University of Central Florida. kpmoran@ucf.edu. 407-823-5239. Research collaborator and mentor.

Elaine Weyuker. University Distinguished Professor. University of Central Florida. Email available upon request. 407-823-3957. Research collaborator and mentor.

Suzanne Rivoire. Professor of Computer Science. Sonoma State University. rivoire@sonoma.edu. 707-664-3337. Research collaborator and mentor.

Peter Goodman. Industry researcher. Current employer is private. Contact information available upon request. Manager and mentor while working at Trail of Bits.